

# WorkFlow Global HRMS - Part 8: Document Management Module

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## 1. Introduction

Part 8 implements the Document Management Module for the WorkFlow Global HRMS system. This module enables employees to upload and manage their employment documents, while allowing HR to verify and track document compliance. The system supports multiple document categories including identity proofs, tax documents, work authorization, and educational certificates.

## 2. Architecture Overview

### 2.1 Module Structure

The Document Management Module follows the established layered architecture:

- DocumentFeature.Domain: Contains Document entity and enums
- DocumentFeature.Application: Business logic, DTOs, and service interfaces
- DocumentFeature.Infrastructure: Repository implementations and DI configuration
- DocumentFeature.GraphQL: Query and mutation resolvers for API exposure

### 2.2 Domain Entity

Document Entity:

```
public class Document : BaseEntity
{
    public string EmployeeId { get; set; }
    public DocumentCategory Category { get; set; }
    public string Name { get; set; }
    public string? Url { get; set; }
    public DateTime? ExpiryDate { get; set; }
    public DocumentStatus Status { get; set; }
    public string? RejectionReason { get; set; }
}
```

### 2.3 Document Categories

The system supports four main document categories:

- Identity: Passport, Driver License, National ID, Aadhaar, PAN Card
- Tax: W-4, W-2, Form 16, IT Returns, Tax Declarations
- Work Authorization: Visa, Work Permit, Employment Authorization
- Education: Degrees, Certificates, Transcripts, Professional Licenses

### 2.4 Document Status Workflow

Documents progress through the following statuses:

- Missing: Document not yet uploaded by employee
- Uploaded: Employee has submitted document, pending HR review
- Verified: HR has reviewed and approved the document
- Rejected: HR has rejected the document with reason

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## 3. Backend Implementation

### 3.1 Repository Layer

IDocumentRepository Interface:

```
public interface IDocumentRepository
{
    Task<IEnumerable<Document>> GetEmployeeDocumentsAsync(string employeeId);
    Task<Document?> GetDocumentByIdAsync(string documentId);
    Task<Document> AddItemAsync(Document document);
    Task<Document> UpdateItemAsync(string id, Document document);
}
```

### 3.2 Application Layer - Document Service

The DocumentService implements core business logic:

```
public interface IDocumentService
{
    Task<IEnumerable<DocumentDto>> GetMyDocumentsAsync(string employeeId);
    Task<DocumentDto?> UploadDocumentAsync(UploadDocumentRequest request);
    Task<DocumentDto?> VerifyDocumentAsync(string documentId);
    Task<DocumentDto?> RejectDocumentAsync(string documentId, string reason);
}
```

### 3.3 Document Upload Flow

When an employee uploads a document:

- 1. Employee selects document category and provides metadata
- 2. File is uploaded to cloud storage (Azure Blob/S3)
- 3. Document record created with Status = Uploaded
- 4. URL to stored file is saved in database
- 5. Optional expiry date is recorded for time-sensitive documents
- 6. HR is notified of new document for review

### 3.4 Document Verification Flow

When HR verifies a document:

- 1. HR reviews uploaded document
- 2. If valid, status updated to Verified
- 3. RejectionReason cleared if previously rejected
- 4. Employee notified of verification
- 5. Document marked as compliant

### 3.5 Document Rejection Flow

When HR rejects a document:

- 1. HR provides rejection reason

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- 2. Status updated to Rejected
- 3. RejectionReason stored for employee reference
- 4. Employee notified with reason
- 5. Employee can re-upload corrected document

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## 4. GraphQL API Layer

### 4.1 Queries

getMyDocuments:

```
[GraphQLName("getMyDocuments")]
public async Task<IEnumerable<DocumentDto>> GetMyDocumentsAsync(
    [Service] IDocumentService documentService,
    string employeeId)
{
    return await documentService.GetMyDocumentsAsync(employeeId);
}
```

### 4.2 Mutations

uploadDocument:

```
[GraphQLName("uploadDocument")]
public async Task<DocumentDto?> UploadDocumentAsync(
    [Service] IDocumentService documentService,
    string employeeId,
    DocumentCategory category,
    string name,
    string url,
    DateTime? expiryDate)
{
    var request = new UploadDocumentRequest
    {
        EmployeeId = employeeId,
        Category = category,
        Name = name,
        Url = url,
        ExpiryDate = expiryDate
    };
    return await documentService.UploadDocumentAsync(request);
}
```

verifyDocument and rejectDocument:

```
[GraphQLName("verifyDocument")]
public async Task<DocumentDto?> VerifyDocumentAsync(
    [Service] IDocumentService documentService,
    string documentId)
{
    return await documentService.VerifyDocumentAsync(documentId);
}

[GraphQLName("rejectDocument")]
public async Task<DocumentDto?> RejectDocumentAsync(
    [Service] IDocumentService documentService,
    string documentId,
```

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```
    string reason)
{
    return await documentService.RejectDocumentAsync(documentId, reason);
}
```

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## 5. Frontend Implementation

### 5.1 Document Repository Dashboard

The Documents Dashboard (/documents) provides:

- Status Cards: Show counts of Missing, Uploaded, Verified, Rejected documents
- Category Filters: Filter by Identity, Tax, Work Auth, Education
- Document List: Display all documents with status badges
- Upload Button: Opens modal for document submission
- View/Download: Access uploaded document files

### 5.2 Document Upload Form

The Upload Document modal includes:

- Category dropdown: Select document type
- Document name: Descriptive name for the document
- File upload: Browse and select file (PDF, JPG, PNG)
- Expiry date: Optional for time-sensitive documents
- Validation: File size limits, allowed formats

### 5.3 HR Document Review Queue

Available for HR role:

- Pending Documents: List of uploaded documents awaiting review
- Employee Information: Name, department, document category
- Document Preview: View uploaded file
- Verify Button: Approve document
- Reject Button: Opens modal for rejection reason
- Bulk Actions: Verify multiple documents at once

### 5.4 Document Status Indicators

Visual status indicators:

- Missing: Red badge with warning icon
- Uploaded: Yellow badge with clock icon
- Verified: Green badge with checkmark icon
- Rejected: Red badge with X icon, shows rejection reason

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## 6. Document Compliance & Tracking

### 6.1 Required Documents by Country

India:

- Identity: Aadhaar Card, PAN Card
- Tax: Form 16, IT Returns
- Education: Degree Certificates, Mark Sheets
- Work: Offer Letter, Appointment Letter

United States:

- Identity: Social Security Card, Driver License
- Tax: W-4, W-2 (previous employer)
- Work Authorization: I-9, Visa/Work Permit (if applicable)
- Education: Degree Certificates, Transcripts

### 6.2 Expiry Tracking

System tracks document expiration:

- Expiry date stored for time-sensitive documents
- Automated reminders 30/60/90 days before expiry
- Dashboard shows expiring documents
- HR notified of expired documents
- Employee blocked from certain actions if critical docs expired

### 6.3 Compliance Dashboard

HR compliance view shows:

- Overall compliance percentage by department
- Missing documents by employee
- Pending verifications count
- Expiring documents in next 90 days
- Rejected documents requiring resubmission



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## 7. Security & Privacy Considerations

### 7.1 Access Control

Document access rules:

- Employees can only view/upload their own documents
- HR can view all employee documents
- Managers cannot view team member documents (privacy)
- Document URLs are signed with expiration
- Audit log tracks all document access

### 7.2 Data Protection

Security measures:

- Documents encrypted at rest in cloud storage
- Encrypted in transit using HTTPS/TLS
- Access tokens expire after 1 hour
- No direct file system access
- Virus scanning on upload
- Watermarking for sensitive documents

### 7.3 Compliance

Regulatory compliance:

- GDPR: Right to access, right to deletion
- Data retention: Documents retained per legal requirements
- Audit trail: Track who accessed which document when
- Consent: Employee consent for document storage
- Data portability: Export documents on request

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## 8. Technical Interview Questions

### 8.1 Architecture Questions

Q1: Why store document URL instead of binary data in database?

A: Storing URLs provides: 1) Better scalability (database not bloated), 2) Faster queries, 3) Leverage cloud storage features (CDN, versioning), 4) Cost-effective storage, 5) Easy integration with cloud services.

Q2: How would you handle document versioning?

A: Add Version field to Document entity. When employee re-uploads, create new record with incremented version, mark old version as superseded. Maintain history for audit. HR verifies latest version only.

### 8.2 Security Questions

Q3: How do you prevent unauthorized document access?

A: 1) Generate signed URLs with expiration, 2) Validate user permissions before generating URL, 3) Use cloud storage access policies, 4) Implement rate limiting, 5) Log all access attempts, 6) Use separate storage buckets per tenant.

Q4: How would you implement document encryption?

A: Use cloud provider encryption (Azure Storage SSE, S3 SSE). For additional security: 1) Client-side encryption before upload, 2) Store encryption keys in Key Vault, 3) Use envelope encryption, 4) Rotate keys periodically.

### 8.3 Performance Questions

Q5: How would you optimize document upload for large files?

A: 1) Implement chunked upload, 2) Use multipart upload for files >5MB, 3) Show progress bar, 4) Resume failed uploads, 5) Compress images before upload, 6) Use CDN for downloads, 7) Implement background processing for virus scanning.

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## 9. Future Enhancements

### 9.1 Advanced Features

- OCR: Extract text from uploaded documents
- Auto-categorization: AI-based document classification
- Smart validation: Verify document authenticity
- E-signature: Digital signing of documents
- Document templates: Pre-filled forms for employees

### 9.2 Integration

- Background verification: Integrate with verification services
- Government APIs: Verify Aadhaar, PAN, SSN
- Email integration: Send documents via email
- Onboarding: Auto-request documents for new hires

### 9.3 Reporting

- Compliance reports: Track document submission rates
- Expiry reports: Upcoming expirations
- Audit reports: Document access logs
- Verification metrics: HR processing times

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## 10. Conclusion

Part 8 successfully implements a comprehensive Document Management Module for the WorkFlow Global HRMS system. The implementation provides secure document storage, verification workflow, and compliance tracking while maintaining strict privacy and security standards.

Key achievements include:

- Secure document upload and storage
- HR verification workflow with rejection handling
- Document categorization and status tracking
- Expiry date monitoring and alerts
- Role-based access control
- Compliance dashboard for HR
- GraphQL API for document operations

The module is production-ready and can be extended with features such as OCR, AI-based classification, e-signatures, and integration with government verification services.